2.24 Differentiation: Inverse Trigonometric Functions

Worked Solutions

1 a)

$$\begin{aligned} y &= \arcsin x \\ \Rightarrow \sin y &= x \\ \frac{d}{dx}(\sin y) &= \frac{d}{dx}(x) \\ \Rightarrow \cos y \frac{dy}{dx} &= 1 \\ \Rightarrow \frac{dy}{dx} &= \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}. \end{aligned}$$

1 b)

$$\begin{aligned} y &= \arccos x \\ \Rightarrow \cos y &= x \\ \frac{d}{dx}(\cos y) &= \frac{d}{dx}(x) \\ \Rightarrow -\sin y \frac{dy}{dx} &= 1 \\ \Rightarrow \frac{dy}{dx} &= -\frac{1}{\sin y} = -\frac{1}{\sqrt{1-\cos^2 y}} = -\frac{1}{\sqrt{1-x^2}}. \end{aligned}$$

1 c)

$$\begin{aligned} y &= \arctan x \\ \Rightarrow \tan y &= x \\ \frac{d}{dx}(\tan y) &= \frac{d}{dx}(x) \\ \Rightarrow \sec^2 y \frac{dy}{dx} &= 1 \\ \Rightarrow \frac{dy}{dx} &= \frac{1}{\sec^2 y} = \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2}. \end{aligned}$$

2 a)

$$\begin{aligned} y = \arcsin x, \qquad \frac{dy}{dx} &= \frac{1}{\sqrt{1-x^2}}, \\ \left.\frac{dy}{dx}\right|_{x=1/3} &= \frac{1}{\sqrt{1-(1/3)^2}} = \frac{1}{\sqrt{8/9}} = \frac{3}{2\sqrt{2}} = \frac{3\sqrt{2}}{4}. \end{aligned}$$

2 b)

$$y = \arccos x, \qquad \frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}},$$
$$\left.\frac{dy}{dx}\right|_{x=-1/4} = -\frac{1}{\sqrt{1-(-1/4)^2}} = -\frac{1}{\sqrt{15/16}} = -\frac{4}{\sqrt{15}} = -\frac{4\sqrt{15}}{15}.$$

2 c)

$$y = \arctan x, \qquad \frac{dy}{dx} = \frac{1}{1+x^2},$$
$$\left.\frac{dy}{dx}\right|_{x=2/5} = \frac{1}{1+(2/5)^2} = \frac{1}{1+4/25} = \frac{1}{29/25} = \frac{25}{29}.$$

3 a)

$$y = \arcsin 2x \quad \Rightarrow \quad \frac{dy}{dx} = 2 \times \frac{1}{\sqrt{1-(2x)^2}} = \frac{2}{\sqrt{1-4x^2}}.$$

3 b)

$$y = \arccos 3x \quad \Rightarrow \quad \frac{dy}{dx} = 3 \times \left(-\frac{1}{\sqrt{1-(3x)^2}}\right) = -\frac{3}{\sqrt{1-9x^2}}.$$

3 c)

$$y = \arctan 8x \quad \Rightarrow \quad \frac{dy}{dx} = 8 \times \frac{1}{1+(8x)^2} = \frac{8}{1+64x^2}.$$

3 d)

$$y = \arcsin \frac{x}{3}$$
$$\Rightarrow \frac{dy}{dx} = \frac{1}{3} \times \frac{1}{\sqrt{1-(x/3)^2}} = \frac{1}{3\sqrt{1-x^2/9}}$$
$$= \frac{1}{3\sqrt{(9-x^2)/9}} = \frac{1}{\sqrt{9-x^2}}.$$

3 e)

$$y = \arccos \frac{x}{5}$$
$$\Rightarrow \frac{dy}{dx} = \frac{1}{5} \times \left(-\frac{1}{\sqrt{1-(x/5)^2}}\right)$$

$$= -\frac{1}{5\sqrt{1-x^2/25}} = -\frac{1}{5\sqrt{(25-x^2)/25}}$$
$$= -\frac{1}{\sqrt{25-x^2}}.$$

3 f)

$$y = \arctan\left(-\frac{2}{3}x\right)$$
$$\Rightarrow \frac{dy}{dx} = -\frac{2}{3} \times \frac{1}{1+(-2x/3)^2} = -\frac{2}{3(1+4x^2/9)}$$
$$= -\frac{2}{3+4x^2/3} = -\frac{6}{9+4x^2}.$$

4 a)

$$\frac{d}{dx}\left(\arcsin\frac{x}{2}\right) = \frac{1}{2} \times \frac{1}{\sqrt{1-(x/2)^2}}$$
$$= \frac{1}{2\sqrt{1-x^2/4}} = \frac{1}{2\sqrt{(4-x^2)/4}}$$
$$= \frac{1}{2 \times \frac{1}{2}\sqrt{4-x^2}} = \frac{1}{\sqrt{4-x^2}}.$$

4 b)

$$\frac{d}{dx}\left(\arccos\frac{3x}{4}\right) = \frac{3}{4} \times \left(-\frac{1}{\sqrt{1-(3x/4)^2}}\right)$$
$$= -\frac{3}{4\sqrt{1-9x^2/16}} = -\frac{3}{4\sqrt{(16-9x^2)/16}}$$
$$= -\frac{3}{\sqrt{16-9x^2}}.$$

4 c)

$$\frac{d}{dx}\left(\arctan\frac{5x}{3}\right) = \frac{5}{3} \times \frac{1}{1+(5x/3)^2}$$
$$= \frac{5}{3+25x^2/3} = \frac{15}{9+25x^2}.$$

5)

$$y = 8\arcsin\frac{x}{4},$$
$$\frac{dy}{dx} = 8 \times \frac{1}{4} \times \frac{1}{\sqrt{1-(x/4)^2}} = \frac{2}{\sqrt{1-x^2/16}},$$
$$\left.\frac{dy}{dx}\right|_{x=2} = \frac{2}{\sqrt{1-(1/16)(2)^2}} = \frac{2}{\sqrt{3/4}} = 2 \times \frac{2}{\sqrt{3}} = \frac{4}{\sqrt{3}} = \frac{4\sqrt{3}}{3}.$$

6)

$$y = 3\arccos x, \qquad \frac{dy}{dx} = -\frac{3}{\sqrt{1-x^2}},$$
$$y\big|_{x=1/2} = 3\arccos\frac{1}{2} = \frac{\pi}{3},$$
$$\left.\frac{dy}{dx}\right|_{x=1/2} = -\frac{3}{\sqrt{1-(1/2)^2}} = -\frac{3}{\sqrt{3/4}} = -2\sqrt{3}.$$

So the tangent is

$$y - \frac{\pi}{3} = -2\sqrt{3}\left(x - \frac{1}{2}\right) \quad \Rightarrow \quad y = -2\sqrt{3}x + \sqrt{3} + \frac{\pi}{3}.$$

7)

$$y = 2\arccos\frac{x}{3} - \arcsin\frac{x}{3},$$
$$\frac{dy}{dx} = -\frac{2}{3\sqrt{1-x^2/9}} - \frac{1}{3\sqrt{1-x^2/9}} = -\frac{1}{\sqrt{1-x^2/9}}.$$

At $x = \frac{3}{\sqrt{2}}$,

$$y = 2\arccos\frac{1}{\sqrt{2}} - \arcsin\frac{1}{\sqrt{2}} = 2\times\frac{\pi}{4} - \frac{\pi}{4} = \frac{\pi}{4},$$
$$\left.\frac{dy}{dx}\right|_{x=3/\sqrt{2}} = -\frac{1}{\sqrt{1-\frac{1}{9}(3/\sqrt{2})^2}} = -\sqrt{2}.$$

So the tangent is $y - \frac{\pi}{4} = -\sqrt{2}(x - \frac{3}{\sqrt{2}})$, hence $y = -\sqrt{2}x + 3 + \frac{\pi}{4}$.

8)

$$y = 4\arctan\frac{2x}{5},$$
$$\frac{dy}{dx} = 4\times\frac{2}{5}\times\frac{1}{1+(2x/5)^2} = \frac{40}{25+4x^2}.$$

At $x = \frac{5}{2}$, $y = 4\arctan 1 = \pi$ and $\frac{dy}{dx} = \frac{40}{25+25} = \frac{4}{5}$. So

$$y - \pi = \frac{4}{5}\left(x - \frac{5}{2}\right) \quad \Rightarrow \quad y = \frac{4}{5}x + \pi - 2.$$

9)

$$y = 4 - 3\arctan\frac{x}{4} - 2\arctan\frac{x}{3},$$
$$\frac{dy}{dx} = -\frac{3}{4}\frac{1}{1+(x/4)^2} - \frac{2}{3}\frac{1}{1+(x/3)^2}$$
$$= -\frac{12}{16+x^2} - \frac{6}{9+x^2}.$$

At $x = 0$, $y = 4$ and $\frac{dy}{dx} = -\frac{12}{16} - \frac{6}{9} = -\frac{17}{12}$. The normal gradient is $\frac{12}{17}$, so

$$y - 4 = \frac{12}{17}(x - 0) \quad \Rightarrow \quad 17y - 68 = 12x \quad \Rightarrow \quad -12x + 17y = 68.$$

10)

$$f(x) = \arctan x \quad \Rightarrow \quad f'(x) = \frac{1}{1+x^2},$$
$$g(x) = \arctan 3x \quad \Rightarrow \quad g'(x) = 3 \times \frac{1}{1+(3x)^2} = \frac{3}{1+9x^2}.$$

$$\begin{aligned} f'(x) = g'(x) &\Rightarrow \frac{1}{1+x^2} = \frac{3}{1+9x^2} \\ &\Rightarrow 1+9x^2 = 3+3x^2 \\ &\Rightarrow 6x^2 = 2 \Rightarrow x^2 = \frac{1}{3} \\ &\Rightarrow x = \pm\frac{1}{\sqrt{3}} = \pm\frac{\sqrt{3}}{3}. \end{aligned}$$

11 a)

$$y = \arcsin\left(\frac{2}{3}x\right),$$
$$\frac{dy}{dx} = \frac{2}{3} \times \frac{1}{\sqrt{1-(2x/3)^2}} = \frac{2}{\sqrt{9-4x^2}}.$$

For $-\frac{3}{2} \le x \le \frac{3}{2}$, $0 \le 9-4x^2 \le 9$, so $0 \le \sqrt{9-4x^2} \le 3$. This means $\frac{dy}{dx} = \frac{2}{\sqrt{9-4x^2}} \ge \frac{2}{3}$ throughout the interval. Thus the gradient is always at least $\frac{2}{3}$ and is never equal to $\frac{1}{2}$.

11 b)

$$y = \arcsin\left(\frac{2}{3}x\right), \qquad \frac{dy}{dx} = \frac{2}{\sqrt{9-4x^2}},$$
$$y\big|_{x=-3\sqrt{2}/4} = \arcsin\left(\frac{2}{3} \times -\frac{3\sqrt{2}}{4}\right) = \arcsin\left(-\frac{\sqrt{2}}{2}\right) = -\frac{\pi}{4}.$$

So $P = (-\frac{3\sqrt{2}}{4}, -\frac{\pi}{4})$.

$$\left.\frac{dy}{dx}\right|_{x=-3\sqrt{2}/4} = \frac{2}{\sqrt{9-4(-3\sqrt{2}/4)^2}} = \frac{2\sqrt{2}}{3}.$$

The normal gradient is $-\frac{3}{2\sqrt{2}}$, so its equation is

$$y + \frac{\pi}{4} = -\frac{3}{2\sqrt{2}}\left(x + \frac{3\sqrt{2}}{4}\right).$$

The normal crosses the y-axis when $x = 0$:

$$y + \frac{\pi}{4} = -\frac{3}{2\sqrt{2}}\left(\frac{3\sqrt{2}}{4}\right) \Rightarrow y = -\frac{9}{8} - \frac{\pi}{4}.$$

Thus $Q = (0, -\frac{9}{8} - \frac{\pi}{4})$, and the area of triangle OPQ is

$$\begin{aligned} \frac{1}{2}\left|-\frac{9}{8}-\frac{\pi}{4}\right|\left|-\frac{3\sqrt{2}}{4}\right| &= \frac{3\sqrt{2}}{8}\left(\frac{9}{8}+\frac{\pi}{4}\right) \\ &= \frac{3\sqrt{2}}{64}(9+2\pi) \text{ units}^2. \end{aligned}$$

12)

$$\begin{aligned} y &= \arcsin x - \arccos x, \\ \frac{dy}{dx} &= \frac{1}{\sqrt{1-x^2}} - \left(-\frac{1}{\sqrt{1-x^2}}\right) \\ &= \frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}} = \frac{2}{\sqrt{1-x^2}}. \end{aligned}$$

For $-1 < x < 1$, $\sqrt{1-x^2} > 0$, so $\frac{2}{\sqrt{1-x^2}} > 0$. Hence the curve is increasing for $-1 < x < 1$.

13 a)

$$\begin{aligned} y &= 3\arctan 6x, \\ \frac{dy}{dx} &= 3 \times 6 \times \frac{1}{1+(6x)^2} = \frac{18}{1+36x^2}. \end{aligned}$$

Set $\frac{dy}{dx} = 24$:

$$\begin{aligned} \frac{18}{1+36x^2} &= 24 \\ 18 &= 24 + 864x^2 \\ x^2 &= -\frac{1}{44}. \end{aligned}$$

This has no real solutions, so there are no points on the curve where the gradient is equal to 24.

13 b)

For all real x, $1 + 36x^2 \geq 1$, with equality when $x = 0$. Therefore the maximum value of $\frac{dy}{dx}$ occurs when $x = 0$, and the maximum gradient is 18.

14)

$$\begin{aligned} y &= \arctan 2x - x^2, \\ \frac{dy}{dx} &= 2 \times \frac{1}{1+(2x)^2} - 2x \\ &= \frac{2}{1+4x^2} - 2x \\ &= \frac{2 - 2x(1+4x^2)}{1+4x^2} \\ &= \frac{2 - 2x - 8x^3}{1+4x^2}. \end{aligned}$$

Stationary points exist when $\frac{dy}{dx} = 0$:

$$\begin{aligned} \frac{2 - 2x - 8x^3}{1+4x^2} &= 0 \\ 2 - 2x - 8x^3 &= 0 \\ 4x^3 + x - 1 &= 0 \end{aligned}$$

$$x = \frac{1}{2}.$$

$$\begin{aligned} y\big|_{x=1/2} &= \arctan\left(2 \times \frac{1}{2}\right) - \left(\frac{1}{2}\right)^2 \\ &= \arctan 1 - \frac{1}{4} = \frac{\pi}{4} - \frac{1}{4} = \frac{1}{4}(\pi - 1). \end{aligned}$$

So there is a single stationary point at $\left(\frac{1}{2}, \frac{1}{4}(\pi - 1)\right)$.

15 a)

$$\begin{aligned} f(x) &= \arctan 3x, \\ f'(x) &= 3 \times \frac{1}{1 + (3x)^2} = \frac{3}{1 + 9x^2}. \end{aligned}$$

15 b)

$$\begin{aligned} f'(x) &= 3(1 + 9x^2)^{-1} \\ &\approx 3\left(1 + (-1)(9x^2) + \frac{(-1)(-2)}{2}(9x^2)^2 + \cdots\right) \\ &= 3 - 27x^2 + 243x^4 + \cdots . \end{aligned}$$

15 c)

$$\begin{aligned} \arctan 3x &\approx \int (3 - 27x^2 + 243x^4 + \cdots)\, dx \\ &= 3x - 9x^3 + \frac{243}{5}x^5 + \cdots . \end{aligned}$$

16)

$$\begin{aligned} y &= p \arcsin x - qx, \\ \frac{dy}{dx} &= \frac{p}{\sqrt{1 - x^2}} - q. \end{aligned}$$

Stationary points exist when $\frac{dy}{dx} = 0$:

$$\begin{aligned} \frac{p}{\sqrt{1 - x^2}} - q &= 0 \\ \frac{p}{\sqrt{1 - x^2}} &= q \\ \sqrt{1 - x^2} &= \frac{p}{q} \\ 1 - x^2 &= \frac{p^2}{q^2} \end{aligned}$$

$$x^2 = 1 - \frac{p^2}{q^2}$$
$$x = \pm\sqrt{1 - \frac{p^2}{q^2}}.$$

If $0 < p < q$ (as p and q are both positive constants), then $p^2 < q^2$, and so $\frac{p^2}{q^2} < 1$. Therefore $1 - \frac{p^2}{q^2} > 0$, and $y = f(x)$ has precisely two stationary points when $p < q$.

17)

$$y = 5x + \sqrt{21}\arccos x,$$
$$\frac{dy}{dx} = 5 - \frac{\sqrt{21}}{\sqrt{1-x^2}}.$$

The curve is increasing when $\frac{dy}{dx} > 0$:

$$5 - \frac{\sqrt{21}}{\sqrt{1-x^2}} > 0$$
$$5 > \frac{\sqrt{21}}{\sqrt{1-x^2}}$$
$$\sqrt{1-x^2} > \frac{\sqrt{21}}{5}$$
$$1 - x^2 > \frac{21}{25}$$
$$0 > x^2 - \frac{4}{25}$$
$$\left(x - \frac{2}{5}\right)\left(x + \frac{2}{5}\right) < 0$$
$$-\frac{2}{5} < x < \frac{2}{5}.$$

The curve is increasing when $\left\{x : x \in \mathbb{R}, -\frac{2}{5} < x < \frac{2}{5}\right\}$.

18)

$$y = 3\arctan\frac{x}{6} - 2\arctan\frac{x}{4},$$
$$\begin{aligned}\frac{dy}{dx} &= 3 \times \frac{1}{6} \times \frac{1}{1 + (x/6)^2} - 2 \times \frac{1}{4} \times \frac{1}{1 + (x/4)^2} \\ &= \frac{18}{36 + x^2} - \frac{8}{16 + x^2} \\ &= \frac{18(16 + x^2) - 8(36 + x^2)}{(36 + x^2)(16 + x^2)} \\ &= \frac{10x^2}{x^4 + 52x^2 + 576}.\end{aligned}$$

Stationary points exist when $\frac{dy}{dx} = 0$:

$$\frac{10x^2}{x^4 + 52x^2 + 576} = 0$$

$$10x^2 = 0$$
$$x = 0.$$

$$y\big|_{x=0} = 3\arctan 0 - 2\arctan 0 = 0 - 0 = 0.$$

So there is a stationary point at $(0, 0)$.

$$\frac{d^2y}{dx^2} = \frac{20x(x^4 + 52x^2 + 576) - 10x^2(4x^3 + 104x)}{(x^4 + 52x^2 + 576)^2}$$
$$= \frac{-20x^5 + 11520x}{(x^4 + 52x^2 + 576)^2}.$$

Points of inflection exist when $\frac{d^2y}{dx^2} = 0$. Since $\frac{d^2y}{dx^2}\Big|_{x=0} = 0$, the origin is a potential stationary point of inflection. Also, $\frac{d^2y}{dx^2}\Big|_{x=-0.1} = -0.00346\ldots < 0$, so the curve is concave (concave downwards) here, and $\frac{d^2y}{dx^2}\Big|_{x=0.1} = 0.00346\ldots > 0$, so the curve is convex (concave upwards) here. As the second derivative changes sign either side of the origin, we must have that the origin is a non-stationary point of inflection.

19 a)

$$y = \arcsin x^2 \quad \Rightarrow \quad \frac{dy}{dx} = 2x \times \frac{1}{\sqrt{1 - (x^2)^2}} = \frac{2x}{\sqrt{1 - x^4}}.$$

19 b)

$$y = \arccos\sqrt{x} = \arccos(x^{1/2}),$$
$$\frac{dy}{dx} = \frac{1}{2}x^{-1/2} \times \left(-\frac{1}{\sqrt{1 - (x^{1/2})^2}}\right)$$
$$= -\frac{1}{2\sqrt{x}\sqrt{1 - x}}$$
$$= -\frac{1}{2\sqrt{x(1 - x)}}.$$

19 c)

$$y = \arctan\frac{1}{x} = \arctan(x^{-1}),$$
$$\frac{dy}{dx} = -x^{-2} \times \frac{1}{1 + (1/x)^2}$$
$$= -\frac{1}{x^2(1 + 1/x^2)}$$
$$= -\frac{1}{x^2 + 1}.$$

19 d)

$$\begin{aligned} y &= \cos(\arcsin 2x), \\ \frac{dy}{dx} &= 2 \times \frac{1}{\sqrt{1-(2x)^2}} \times [-\sin(\arcsin 2x)] \\ &= -\frac{2}{\sqrt{1-4x^2}} \times 2x \\ &= -\frac{4x}{\sqrt{1-4x^2}}. \end{aligned}$$

19 e)

$$\begin{aligned} y &= \sin\left(\arccos\frac{x}{2}\right), \\ \frac{dy}{dx} &= \frac{1}{2} \times \left(-\frac{1}{\sqrt{1-(x/2)^2}}\right) \times \cos\left(\arccos\frac{x}{2}\right) \\ &= -\frac{1}{2\sqrt{1-x^2/4}} \times \frac{x}{2} \\ &= -\frac{x}{4\sqrt{(4-x^2)/4}} \\ &= -\frac{x}{2\sqrt{4-x^2}}. \end{aligned}$$

19 f)

$$\begin{aligned} y &= \cos(\arctan 4x), \\ \frac{dy}{dx} &= 4 \times \frac{1}{1+(4x)^2} \times [-\sin(\arctan 4x)]. \end{aligned}$$

Let $\theta = \arctan 4x$, so $\tan\theta = 4x = \frac{\text{opp}}{\text{adj}}$. Using a right-angled triangle, $\sin\theta = \frac{4x}{\sqrt{(4x)^2+1^2}} = \frac{4x}{\sqrt{16x^2+1}}$, and hence $\theta = \arcsin\left(\frac{4x}{\sqrt{16x^2+1}}\right)$. Therefore

$$\begin{aligned} \frac{dy}{dx} &= 4 \times \frac{1}{1+(4x)^2} \times \left[-\sin\left(\arcsin\frac{4x}{\sqrt{16x^2+1}}\right)\right] \\ &= -\frac{4}{16x^2+1} \times \frac{4x}{\sqrt{16x^2+1}} \\ &= -\frac{16x}{(16x^2+1)^{3/2}}. \end{aligned}$$

20)

$$\begin{aligned} y &= (\arcsin x)^2, \\ \frac{dy}{dx} &= 2 \times \frac{1}{\sqrt{1-x^2}} \times (\arcsin x) \end{aligned}$$

$$= \frac{2\arcsin x}{\sqrt{1-x^2}},$$
$$y\big|_{x=\sqrt{3}/2} = \left(\arcsin\frac{\sqrt{3}}{2}\right)^2 = \left(\frac{\pi}{3}\right)^2 = \frac{\pi^2}{9},$$
$$\left.\frac{dy}{dx}\right|_{x=\sqrt{3}/2} = \frac{2\arcsin(\sqrt{3}/2)}{\sqrt{1-(\sqrt{3}/2)^2}} = \frac{2\times\pi/3}{\sqrt{1/4}} = \frac{4\pi}{3}.$$

So the equation of the tangent is

$$y - \frac{\pi^2}{9} = \frac{4\pi}{3}\left(x - \frac{\sqrt{3}}{2}\right) \quad \Rightarrow \quad y = \frac{4\pi}{3}x + \frac{\pi^2}{9} - \frac{2\pi\sqrt{3}}{3}.$$

21)

$$y = \arctan(e^x),$$
$$\frac{dy}{dx} = e^x \times \frac{1}{1+(e^x)^2} = \frac{e^x}{1+e^{2x}},$$
$$y\big|_{x=-\frac{1}{2}\ln 3} = \arctan\left(e^{-\frac{1}{2}\ln 3}\right) = \arctan\left(e^{\ln(3^{-1/2})}\right) = \arctan\frac{1}{\sqrt{3}} = \frac{\pi}{6},$$
$$\left.\frac{dy}{dx}\right|_{x=-\frac{1}{2}\ln 3} = \frac{1/\sqrt{3}}{1+1/3} = \frac{\sqrt{3}}{4}.$$

So the equation of the tangent is

$$y - \frac{\pi}{6} = \frac{\sqrt{3}}{4}\left(x + \frac{1}{2}\ln 3\right) \quad \Rightarrow \quad y = \frac{\sqrt{3}}{4}x + \frac{\sqrt{3}}{8}\ln 3 + \frac{\pi}{6}.$$

22 a)

$$y = x\arcsin x,$$
$$\frac{dy}{dx} = x \times \frac{1}{\sqrt{1-x^2}} + 1 \times \arcsin x$$
$$= \frac{x}{\sqrt{1-x^2}} + \arcsin x.$$

22 b)

$$y = x^2\arccos 2x,$$
$$\frac{dy}{dx} = x^2 \times \left(2 \times -\frac{1}{\sqrt{1-(2x)^2}}\right) + 2x \times \arccos 2x$$
$$= -\frac{2x^2}{\sqrt{1-4x^2}} + 2x\arccos 2x.$$

22 c)

$$\begin{aligned} y &= x^3 \arctan(x^2), \\ \frac{dy}{dx} &= x^3 \times \left(2x \times \frac{1}{1+(x^2)^2}\right) + 3x^2 \times \arctan(x^2) \\ &= \frac{2x^4}{1+x^4} + 3x^2 \arctan(x^2). \end{aligned}$$

23)

$$\begin{aligned} y &= (x-5)\sqrt{-x^2+10x-24} + \arcsin(x-5) \\ &= (x-5)(-x^2+10x-24)^{1/2} + \arcsin(x-5), \\ \frac{dy}{dx} &= (x-5) \times \frac{1}{2} \times (-2x+10) \times (-x^2+10x-24)^{-1/2} \\ &\quad + \sqrt{-x^2+10x-24} + \frac{1}{\sqrt{1-(x-5)^2}} \\ &= (x-5) \times \frac{1}{2} \times [-2(x+5)] \times (-x^2+10x-24)^{-1/2} \\ &\quad + \sqrt{-x^2+10x-24} + \frac{1}{\sqrt{-x^2+10x-24}} \\ &= -\frac{(x-5)^2}{\sqrt{-x^2+10x-24}} + \sqrt{-x^2+10x-24} + \frac{1}{\sqrt{-x^2+10x-24}} \\ &= \frac{1-(x-5)^2}{\sqrt{-x^2+10x-24}} + \sqrt{-x^2+10x-24} \\ &= \frac{-x^2+10x-24}{\sqrt{-x^2+10x-24}} + \sqrt{-x^2+10x-24} \\ &= 2\sqrt{-x^2+10x-24}. \end{aligned}$$

So $k = 2$.

24)

$$\begin{aligned} y &= 3x\sqrt{4-x^2} + 12\arctan\left(\frac{x}{\sqrt{4-x^2}}\right) \\ &= 3x(4-x^2)^{1/2} + 12\arctan\left(\frac{x}{(4-x^2)^{1/2}}\right), \\ \frac{dy}{dx} &= 3x \times \frac{1}{2} \times (-2x)(4-x^2)^{-1/2} + 3(4-x^2)^{1/2} \\ &\quad + 12 \times \frac{(4-x^2)^{1/2} - x \times \frac{1}{2} \times (-2x)(4-x^2)^{-1/2}}{4-x^2} \times \frac{1}{1+\left(\frac{x}{(4-x^2)^{1/2}}\right)^2} \\ &= -3x^2(4-x^2)^{-1/2} + 3(4-x^2)^{1/2} \\ &\quad + \frac{12(4-x^2)^{1/2} + 12x^2(4-x^2)^{-1/2}}{4-x^2} \times \frac{1}{1+\frac{x^2}{4-x^2}} \\ &= -3x^2(4-x^2)^{-1/2} + 3(4-x^2)^{1/2} \end{aligned}$$

$$
\begin{aligned}
&+\frac{12(4-x^2)^{1/2}+12x^2(4-x^2)^{-1/2}}{4-x^2}\times\frac{4-x^2}{4-x^2+x^2}\\
&=-3x^2(4-x^2)^{-1/2}+3(4-x^2)^{1/2}+3(4-x^2)^{1/2}+3x^2(4-x^2)^{-1/2}\\
&=6(4-x^2)^{1/2}.
\end{aligned}
$$

So $k=6$.

25)

$$
\begin{aligned}
\arccos y-\arccos 3x&=\frac{\pi}{3},\\
\frac{d}{dx}(\arccos y-\arccos 3x)&=\frac{d}{dx}\left(\frac{\pi}{3}\right),\\
\frac{1}{\sqrt{1-y^2}}\frac{dy}{dx}-3\times\frac{1}{\sqrt{1-(3x)^2}}&=0,\\
\frac{1}{\sqrt{1-y^2}}\frac{dy}{dx}&=\frac{3}{\sqrt{1-9x^2}},\\
\frac{dy}{dx}&=\frac{3\sqrt{1-y^2}}{\sqrt{1-9x^2}}.
\end{aligned}
$$

When $x=\frac{1}{6}$:

$$
\begin{aligned}
\arccos y-\arccos\left(3\times\frac{1}{6}\right)&=\frac{\pi}{3},\\
\arccos y-\arccos\frac{1}{2}&=\frac{\pi}{3},\\
\arccos y-\frac{\pi}{3}&=\frac{\pi}{3},\\
\arccos y&=\frac{2\pi}{3},\\
y&=\cos\frac{2\pi}{3}=-\frac{1}{2}.
\end{aligned}
$$

$$
\left.\frac{dy}{dx}\right|_{(1/6,-1/2)}=\frac{3\sqrt{1-(-1/2)^2}}{\sqrt{1-9(1/6)^2}}=\frac{3\sqrt{3}/2}{\sqrt{3}/2}=3.
$$

26 a)

$$
\begin{aligned}
y&=\operatorname{cosec}^{-1}x,\\
\operatorname{cosec}y&=x,\\
\frac{d}{dx}(\operatorname{cosec}y)&=\frac{d}{dx}(x),\\
-\operatorname{cosec}y\cot y\frac{dy}{dx}&=1,\\
\frac{dy}{dx}&=-\frac{1}{\operatorname{cosec}y\cot y}=-\frac{1}{\operatorname{cosec}y\sqrt{\operatorname{cosec}^2y-1}}\\
&=-\frac{1}{\sqrt{x^2}\sqrt{x^2-1}}=-\frac{1}{\sqrt{x^4-x^2}}.
\end{aligned}
$$

26 b)

$$\begin{aligned} y &= \sec^{-1} x, \\ \sec y &= x, \\ \frac{d}{dx}(\sec y) &= \frac{d}{dx}(x), \\ \sec y \tan y \frac{dy}{dx} &= 1, \\ \frac{dy}{dx} &= \frac{1}{\sec y \tan y} = \frac{1}{\sec y \sqrt{\sec^2 y - 1}} \\ &= \frac{1}{\sqrt{x^2}\sqrt{x^2-1}} = \frac{1}{\sqrt{x^4 - x^2}}. \end{aligned}$$

26 c)

$$\begin{aligned} y &= \cot^{-1} x, \\ \cot y &= x, \\ \frac{d}{dx}(\cot y) &= \frac{d}{dx}(x), \\ -\operatorname{cosec}^2 y \frac{dy}{dx} &= 1, \\ \frac{dy}{dx} &= -\frac{1}{\operatorname{cosec}^2 y} = -\frac{1}{1 + \cot^2 y} = -\frac{1}{1+x^2}. \end{aligned}$$